

PROJECT ANALYSIS
06/2025

UEH
UNIVERSITY

Trường Kinh tế, Luật
và Quản lý Nhà nước
KHOA KINH TẾ

PROJECT PROPOSAL SHARED KITCHEN

SUPERVISOR: TH.S NGUYỄN THỊ TÂM HIỀN



FINAL REPORT

GROUP MEMBER EXECUTION
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Work Distribution and Task Completion Assessment Table

Full Name	Student ID	Assigned Tasks	Contribution (%)	Completion Status
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Mộc Đông Nhi	31231020361	• Project introduction • Equipment seeking • Human resource analysis, financial resources • Overall conclusion	20%	Completed
Đỗ Hồng Phúc	31231025123	• Equipment planning • Financial modeling • Project assessment criteria • Risk simulation	20%	Completed
Huỳnh Tài Đức	31231023810	• Market need analysis • Necessity of the project, • Location & legal context • Implementation plan	20%	Completed
Phạm Hữu Anh Toàn	31231023715	• Market need analysis • Equipment seeking, technology implementation • Recommendation	20%	Completed

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I. INTRODUCING THE PROJECT

Shared kitchen is a business model designed to support food startups, small-scale producers, home-based cooks, and delivery-focused F&B (Food and Beverage) businesses. Our project (Shared Kitchen 4.0) responds to the increasing demand for cost-effective, legally compliant, and fully equipped commercial kitchen spaces in urban settings.

The model operates by leasing a large commercial property and subdividing it into multiple fully functional kitchen units, organized into different segments to accommodate the diverse needs of F&B businesses. Each unit is thoughtfully designed to meet specific operational requirements - whether for cloud kitchens, or small-scale food production. The project's primary source of revenue is the rental income generated from these kitchen units, available on a monthly basis or under customizable contracts.

Shared Kitchen 4.0 eliminates the need for significant upfront investment in kitchen infrastructure. This flexible model enhances convenience and scalability, particularly for entrepreneurs entering the food industry or expanding into the online delivery market.

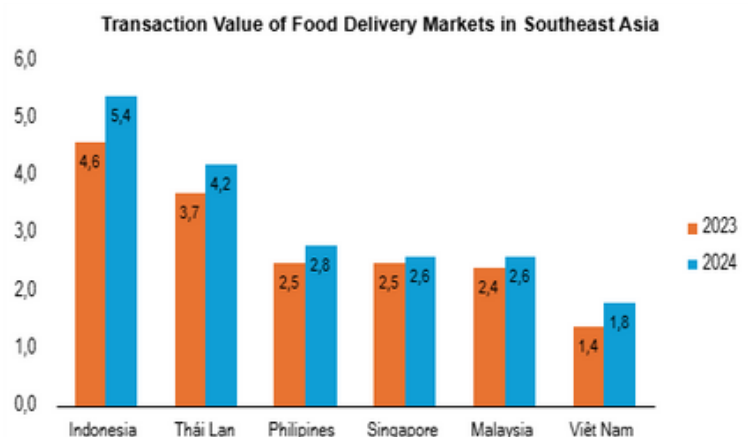
II. NECESSITY OF THE SERVICE

1. Economic and Social Overview & Industry Trends

Vietnam's food and beverage (F&B) industry is experiencing strong recovery post-pandemic, marked by the rapid emergence of small brands, household-run businesses, and food startups. From 2020 to 2024, revenue from foodservice chains and independent outlets increased by 21.5% and 16.2%, respectively—showing significant market momentum.

Consumer demand is shifting rapidly toward delivery-based, convenient, and food safety-focused solutions. According to an IPOS survey, 58.4% of SMEs plan to expand operations despite rising costs—up from 51.7% in 2023.

In 2024, Vietnam's food delivery sector is expected to reach a market value of \$1.8 billion USD, with a 26% growth rate, the highest in Southeast Asia. Despite this, Vietnam still ranks sixth in the region, trailing behind markets like Malaysia, Singapore, and Indonesia - indicating substantial room for growth in this space (Source: Momentum Works).



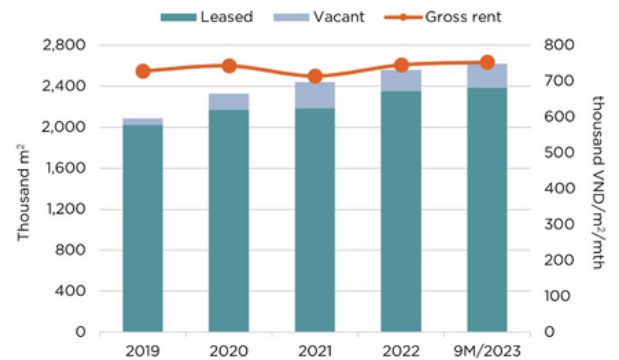
(Source: Momentum Works)

Back in 2020, the COVID-19 pandemic profoundly disrupted Vietnam's urban economy, particularly in Ho Chi Minh City. As dine-in businesses were forced to close, online food delivery surged, and businesses quickly adapted. Cloud kitchens emerged as a vital solution during this time and continue to thrive as online food ordering becomes a habitual behavior for consumers.

2. Rising Rental Costs Drive Demand for Flexible Kitchen Spaces

Rental cost is among the largest operational expenses for F&B businesses in urban areas. Since the 2020 pandemic, commercial rent in cities like HCMC and Hanoi has surged, with prime locations costing between 200 to 800 million VND/month

According to Savills Research & Consultancy, rent now accounts for 25–55% of business revenue, 2–3 times higher than global benchmarks. This puts immense pressure on both large and small F&B operators. For example, a 100 m² retail space on Nguyen Hue Walking Street costs 350–500 million VND/month, while gross margins average below 10%, making profitability difficult. (Source: Savills & Cafef)



(Source: Savills & Cafef)

This landscape is pushing businesses to rethink operations—especially in terms of real estate, staffing, and raw material management—and explore more adaptive models that reduce fixed costs and improve scalability.

3. Policy Support Context and Food Safety Trends

In a recent National Assembly session on Resolution 68, a delegate highlighted the land-access challenge for SMEs:

“Thousands of small and medium-sized businesses are looking to expand production, but suitable land is unavailable. Industrial zones and rental warehouses are typically 1,000 m² or larger, priced far beyond SMEs’ capabilities.”

In response, the Vietnamese government continues to promote SME development through policies such as the Law on Supporting Small and Medium Enterprises (2017), preferential loan programs, and digital transformation initiatives. These policies encourage cost-efficient, flexible business models that leverage shared resources—making the shared kitchen model highly aligned with national development goals

Additionally, as living standards rise, consumer awareness of food safety and hygiene has become more prominent. Shared kitchens must therefore comply with strict food safety laws, creating not only a physical space but also a regulatory-compliant, scalable infrastructure for modern food enterprises.

Moreover, with the rising standard of living, people are becoming increasingly concerned about food safety, especially in the context of a series of food hygiene and safety violations. In group-based models, the shared kitchen is considered a "food service business establishment" or a "food production and trading facility," and is therefore required to fully comply with infrastructure standards and safe cooking procedures to prevent cross-contamination. The facility must obtain a Certificate of Eligibility for Food Safety (according to Article 36), which includes documentation such as a business registration certificate, a statement explaining the facility's infrastructure, a health certificate for the owner, and a food safety training record for the manager.

Once certified, the facility is subject to periodic (every 6–12 months) and ad hoc inspections by district-level or health authorities. The shared kitchen is also required to maintain a hygiene log and inspection records. The rental contract must clearly specify that the tenant is responsible for sourcing ingredients, while the kitchen owner manages equipment, sanitation, and support for food safety procedures to prevent disputes

Key Provisions from Law 55/2010/QH12 that applied with shared kitchen models

- **Article 18:** Packaging and containers must be safe and standards-compliant.
- **Article 19:** Facilities must have an appropriate location and size; clean water; washing, sterilization, and waste-treatment equipment; required documentation; and certified personnel.
- **Article 20:** Storage areas must be spacious enough to deter pests and provide temperature and light control.
- **Article 22:** Small-scale facilities must maintain safe distances from contaminants, ensure clean water, use proper equipment, regulate chemicals, collect waste, and keep records.
- **Article 28:** Food preparation and service zones must separate “raw” and “cooked” areas; provide clean water; install waste-collection, lighting, ventilation, handwashing stations, and proper storage.
- **Article 29:** Utensils for raw and cooked food must be separate and safe; dining utensils must be clean and dry; staff must meet health and food-safety training requirements.

In practice, shared-kitchen operators should also reference implementing Decrees and Circulars—such as Decree 15/2018/NĐ-CP (guiding the Food Safety Law) and Circular 47/2014/TT-BYT (detailing facility conditions)—to ensure full compliance

4. Solution

4.1. Cost Efficiency Amid Rising Rents

In major cities like Ho Chi Minh City, commercial rental prices for F&B spaces have surged—ranging from 200 to 800 million VND/month, often consuming 25–55% of total revenue (Savills, 2023). This presents a major barrier for SMEs and startups.

The shared kitchen model offers a low-cost alternative, allowing tenants to rent 10–20 m² at a fraction of the cost (15–30 million VND/month), while sharing high-cost infrastructure like cold storage, ventilation, and waste systems. Rent is flexible (by hour or usage), helping businesses scale efficiently.

4.2. Built-in Food Safety Compliance

Shared kitchens are classified as food production facilities and must meet hygiene regulations under Vietnam’s Food Safety Law (55/2010/QH12). These include:

- Separated “raw” and “cooked” zones
- Sanitation systems, cold-chain storage, proper drainage
- Food Safety Certificate certification for both operator and tenants

The operator handles infrastructure and periodic inspections, while tenants manage ingredients and processing—ensuring compliance without overburdening each business.

4.3. Faster Integration with Delivery Platforms

The model supports direct integration with GrabFood, ShopeeFood, and BeFood, by providing:

- Pre-certified documents
- Standard hygiene logs and layouts
- Support in applying for licenses

This allows tenants to launch faster, increase customer trust, and comply with platform regulations.

III. INVESTMENT PARAMETERS / IMPLEMENTATION PLAN

1. Site and Market Analysis

1.1. Site Selection and Location Potential Assessmet

1.1.1. Location Selection Criteria

To ensure the model meets its commercial objectives, locations were selected based on four key criteria. First, land rental prices had to be affordable, ranging from 70 to 100 million VND per month for spaces between 800 to 1,000 square meters. Second, areas with high population size and density were prioritized to maximize potential customer reach. Third, the team considered the socioeconomic background of the target customers, giving preference to locations with higher poverty rates where affordable food services are in greater demand. Lastly, the presence of active business activity was essential, measured by the number of local markets and the population served per market, to ensure strong commercial potential and customer flow.

After conducting the data, we have the table followed our criteria

Area	Scale of area (km2)	Population scale	Population density (people/km2)	Poverty Rate (%)	Number of marktet	Population Served per Market (%)	Price of land (800-1000m2)
District 1	7,73	205800	26543	1,31	6	34300	unavailable
District 3	4,92	196433	39925	0,21	4	49108,25	unavailable
District 4	4,18	203060	48578	1,29	7	29008,57143	unavailable
District 5	4,27	187510	43913	-	13	14423,84615	unavailable
District 6	7,19	258945	36014	0,72	9	28771,66667	70
District 7	35,69	324620	9095	0,34	9	36068,88889	88-120
District 8	19,18	451290	23529	0,8	17	26546,47059	48 - 148
District 10	5,72	372450	65113	0,11	6	62075	> 180
District 11	5,14	332536	64695	0,66	4	83134	unavailable
District 12	52,78	520175	9855	0,23	9	57797,22222	70 - 100
Bình Thạnh	20,76	490618	23632	0,61	11	44601,63636	unavailable
Gò Vấp	19,74	663313	33602	0,41	8	82914,125	80
Phú Nhuận	4,88	182477	37392	0,33	6	30412,83333	unavailable
Tân Bình	22,38	470350	21016	0,41	4	117587,5	120 - 160
Tân Phú	16,06	464493	28922	0,15	14	33178,07143	70 - 75
Bình Tân	51,89	702650	13541	0,05	6	117108,3333	70 - 100

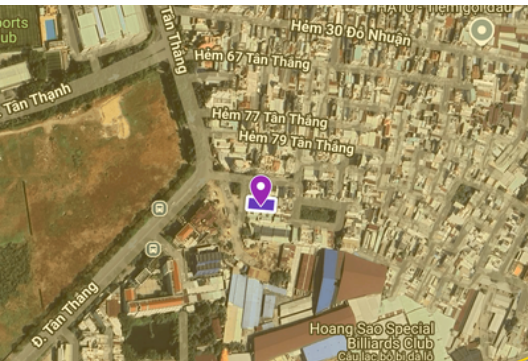
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One key requirement for selecting locations was that the population density exceeded the average of the nine districts under review, ensuring access to a broad and active customer base. After thoroughly assessing market attractiveness and operational feasibility, three districts were identified as the most viable options: District 6, Go Vap District, and Tan Phu District. These areas present strong conditions for the successful implementation of affordable food service models.

1.1.2. Final Site Selection

Comparable land lots in Tan Phu District often rent for 100 million VND per month or more, despite offering smaller frontages, limited flexibility for commercial adaptation, or being situated on narrower streets. Other areas like District 6 or Go Vap District may vary, rental prices tend to fluctuate significantly and often involve trade-offs in terms of accessibility, visibility, or surrounding community density.



Based on comprehensive field surveys and comparative analysis, the team selected the land lot on Tan Thang Street in Son Ky Ward, Tan Phu District. This site stood out due to its strong commercial potential and favorable demographic profile. Son Ky Ward has approximately 48,113 residents, with a population density of 21,000 people per square kilometer—nearly four times the city average. The area features long-standing and stable residential communities, making it an ideal setting for service-based initiatives like the shared kitchen model.



Major residential developments in the vicinity offer a consistent source of both B2B and B2C customers. Celadon City, a large-scale township featuring mid- to high-end housing, is home to nearly 30,000 residents. Nearby, Tan Son Ky Apartments comprises over 662 units, accommodating approximately 2,000 residents. These communities contribute to a stable and accessible customer base for the shared kitchen model.

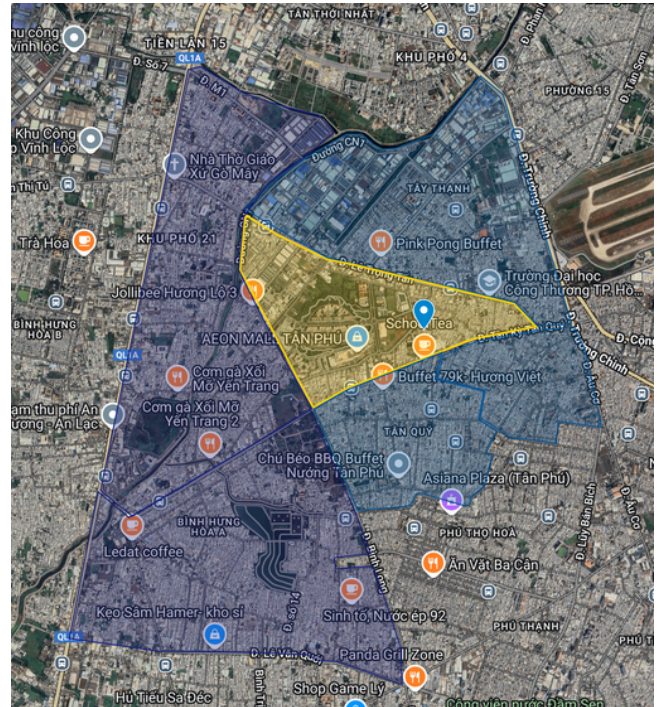
1.1.3. Land Rental Details

After considering multiple options, we selected a land plot measuring 880 m² (16m x 55m), with a 16-meter frontage and located directly across from Celadon City in a vibrant, high-traffic commercial area surrounded by recreational facilities. The agreed rental rate for Year 0 is set at 75 million VND per month, payable at the end of each month, along with a two-month security deposit. Starting from Year 1, the rent will be adjusted annually based on the nominal inflation rate.

[Land rental details on website](#)

1.1.4. Surrounding Areas and Customer Catchment

The selected site on Tan Thang Street presents a compelling combination of accessibility, population density, and cost-effectiveness. Positioned near several densely populated and well-connected wards in Tan Phu and neighboring Binh Tan District, the location benefits from a substantial residential base. To the west, Tay Thanh Ward has over 64,000 residents and key residential clusters like Tay Thanh Apartments and Ruby Garden. Southwest lies Tan Son Nhi Ward, with a high population density of over 37,000 people/km² and developments such as Melody Residences and An Gia Garden. Directly south, Tan Quy Ward is one of the most densely populated in the district that hosts major projects like Carillon 5 and RichStar.



For more detail: [Project location and surrounding areas](#)

Additionally, bordering areas in Binh Tan District, including Binh Hung Hoa A and Binh Hung Hoa, contribute a combined population of over 220,000. Strong road connectivity via Truong Chinh, Le Trong Tan, and Tan Ky Tan Quy further enhances access. With its central location, strong customer base, and favorable rental terms, the Tan Thang Street site offers an ideal foundation for developing a large-scale shared kitchen and F&B logistics hub that supports the growth of emerging food businesses in Ho Chi Minh City.

2. Project Resources

2.1. Equipments

2.1.1. Micro kitchen

For the micro kitchen (3x3 m), the space is designed primarily for storing fresh ingredients, frozen goods and prepared foods; processing and preparing dough alongside other ingredients; as well as organizing utensils and supplies. Specifically, the kitchen is equipped with the following main items:

- **One two-door upright freezer-refrigerator** by Toshiba, capacity 555 L, priced at 15,440,000 VND per unit.
- **One cooler table** by Hoshizaki, capacity 199 L, priced at 29,200,000 VND per unit.
- **One small dough mixer** by Goodfor, capacity 15 L, priced at 7,590,000 VND per unit.
- **One single sink + stainless steel table** by Nam Việt, sink capacity 70 L, priced at 3,850,000 VND per set.
- **Two ingredient shelves** (5 tiers each) by Hidavico, priced at 3,000,000 VND per shelf (total 6,000,000 VND for two).

The total cost of equipment and machinery amounts to 62,080,000 VND for a micro kitchen.

2.1.2. Small kitchen

For the small kitchen (4×5 m), the space is designed primarily for storing larger quantities of ingredients; cooking, baking and frying a variety of dishes; washing and preparing raw materials; as well as organizing utensils and supplies. Specifically, the kitchen is equipped with the following main items:

- **One freezer** by Kenshin, capacity 1,055 L, priced at 35,760,000 VND per unit.
- **One cooler table** by Kenshin, capacity 405 L, priced at 24,560,000 VND per unit.
- **One small convection oven** by Southstar, capacity 4 trays (440×320 mm), priced at 19,800,000 VND per unit.
- **One extractor hood & double sink** by Samsung & Yến Nam, dimensions 500 × 500 × 300 mm, priced at 16,390,000 VND per set.
- **One double fryer (4-burner)** by Goodfor, capacity 12 L, priced at 1,190,000 VND per unit.
- **One European stove** by Thiên Bình, with 4 burners, priced at 7,180,000 VND per unit.
- **Two ingredient shelves (5 tiers each)** by Hidavico, priced at 3,000,000 VND per shelf (total 6,000,000 VND for two).

The total cost of equipment and machinery amounts to 110,880,000 VND for the small kitchen.

2.1.3. Medium kitchen

For the medium kitchen (5×6 m), the area is arranged to carry out all stages of food preparation, cooking, and cleaning according to professional standards, in order to optimize both time and labor. This setup also allows the kitchen to operate like a food business, enabling meal production and delivery through food delivery apps. Thanks to the modern equipment, the quality of dishes is consistently maintained and food safety is assured:

- **One freezer & cooler** by Kenshin, capacity 1,055 L, priced at 71,520,000 VND per unit.
- **Two European stove + fryer + grill units** by Berjaya, dimensions 900 × 790 × 1,070 mm, each priced at 41,326,000 VND per unit.
- **One vacuum packing machine** by Alpha, priced at 5,200,000 VND per unit.
- **Two semi-automatic dishwashers** by Ultrasonic, capacity 1,105 L, each priced at 44,060,000 VND per unit.
- **One extractor hood & sink** by Samsung, priced at 10,990,000 VND per set.

The total cost of equipment and machinery amounts to 258,482,000 VND for the medium kitchen.

2.1.4. Large kitchen

For the large kitchen (10×6 m), the space is arranged to enable large-scale storage, high-capacity cooking, professional packaging, continuous production flow, and efficient ingredient management. It is well-suited for preparing large volumes of meals for schools, companies, hospitals, and other institutions with high meal demands. With those machines:

- **Two freezers, refrigerators, and small cooling storage units** by Kenshin, each with a capacity of 1,630 L, priced at 39,960,000 VND per unit.
- **Two European stove + fryer + grill units** by Berjaya, each measuring 900 × 790 × 1,070 mm, priced at 41,326,000 VND per unit.
- **Two vacuum packing machines** by Bigstar, each measuring 460 × 450 × 1,100 mm, priced at 22,800,000 VND per unit.
- **Two conveyors** by Maruyasu, each 1 m in length, priced at 38,491,200 VND per unit.
- **Six ingredient shelves (5 tiers each)** by Hidavico, priced at 3,000,000 VND per shelf (total 18,000,000 VND for six).

The total cost of equipment and machinery amounts to 303,154,400 VND for the large kitchen.

2.1.5. Other equipments

For efficient handling and transportation of food supplies, the following additional equipment is provided:

- **One Docker lever** by Tân Phát, capacity 6 tấn, priced at 4 275 000 VND per unit (quantity: 1).
- **Five dollies** by Phong Thạnh, capacity 300 kg each, priced at 1 900 000 VND per unit (quantity: 5).

The total cost for these additional handling and transport devices amounts to 13,775,000 VND.

2.1.6. Equipment table breakdown

Type of kitchen	Price per unit	Quantity	Price
Micro kitchen			
2 door upright freezer-refrigerator	15,440,000	1	15,440,000
Cooler table	29,200,000	1	29,200,000
Small dough mixer	7,590,000	1	7,590,000
Single sink + stainless steel table	3,850,000	1	3,850,000
Ingredient shelf	3,000,000	2	6,000,000
Total			62,080,000
Small kitchen			
Freezer	35,760,000	1	35,760,000
Cooler table	24,560,000	1	24,560,000
Small convection oven	19,800,000	1	19,800,000
Extractor hood & double sink	16,390,000	1	16,390,000
Double fryer	1,190,000	1	1,190,000
4 burner European stove	7,180,000	1	7,180,000
Ingredient shelf	3,000,000	2	6,000,000
Total			110,880,000
Medium kitchen			
Freezer & Cooler	71,520,000	1	71,520,000
European stove + fryer + grill	41,326,000	2	82,652,000
Vacuum packing machine	5,200,000	1	5,200,000
Semi-automatic dishwasher	44,060,000	2	88,120,000
Extractor hood & Sink	10,990,000	1	10,990,000
Total			258,482,000
Large kitchen			
Freezer, refrigerator, small cooling storage	39,960,000	2	79,920,000
European stove + fryer + grill	41,326,000	2	82,652,000
Vacuum packing machine	22,800,000	2	45,600,000
Conveyor	38,491,200	2	76,982,400
Ingredient shelf	3,000,000	6	18,000,000
Total			303,154,400
Other equipment			
Docker lever	4,275,000	1	4,275,000
Dolly	1,900,000	5	9,500,000
Total			13,775,000
Equipment cost			748,371,400

2.1.5. Other equipments

For efficient handling and transportation of food supplies, the following additional equipment is provided:

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- **Five dollies** by Phong Thạnh, capacity 300 kg each, priced at 1 900 000 VND per unit (quantity: 5).

The total cost for these additional handling and transport devices amounts to 13,775,000 VND.

2.2. Technology

The following table provides a detailed compilation of the technical solutions deployed to optimize every aspect of operations—ranging from tenant management, invoicing, and equipment monitoring to maintenance, sanitation, and security. Each technology has been selected based on its intended purpose, key features, and flexible tool integrations, thereby ensuring a seamless experience for both kitchen owners and tenants:

No	Technology	Purpose	Key Features	Recommended Tools (Free/Freemium)
1	Tenant & Contract Management	Manage tenant info, contracts, renewals & internal chat	• Centralized tenant database • Contract storage & versioning • Automated renewal reminders • In-app messaging	• Notion (Free Personal) • Google Sheets + AppSheet (Free tier)
2	Billing & Invoicing	Auto-create & send invoices, track payments & reminders	• Invoice templates • Recurring & one-off invoice generation • Payment status dashboard • Email/SMS reminders	• Wave Accounting (100% free) • PayPal (transaction fees only)
3	Operations & Compliance Logging	Log kitchen operations, food-safety inspections & access	• Customizable form builder • Timestamped entries • Auto-reminder notifications • Summary reports & exports	• Google Forms + Sheets (templates + optional Apps Script) • Trello (workflow tracking, optional)
4	Delivery Coordination	Sync orders, update statuses & send notifications	• Multi-platform order ingestion • Real-time status updates • Automated notification workflows • Basic order analytics	• Zapier Free (100 tasks/month) + Google Sheets

2.3. Human resources

One General Administration cum Operations Manager will be recruited with a monthly salary of 18,000,000 VND. This role oversees all administrative and operational activities of the shared kitchen center, including managing lease contracts, ensuring timely rent collection, coordinating internal and external communication, quality control, standardizing procedures, and handling payroll for support staff (security, maintenance, mechanics, and janitorial teams). The manager is also required to possess sufficient knowledge to read and interpret legal documents and business licenses, ensuring that the center operates in full compliance with regulatory requirements.

Two Maintenance Technicians/Mechanics will be employed at a monthly salary of 10,000,000 VND each to ensure the effective operation of all technical systems within the shared kitchen facility. Their responsibilities include maintaining the smooth operation of the entire rental area, repairing and providing backup for malfunctioning equipment, and operating key utility systems such as the electricity room, water purification systems, and waste treatment facilities. In addition, they will oversee daily technical operations including the lighting system, fire safety equipment, cargo delivery, and dock management.

To maintain cleanliness and hygiene, especially critical in food-related environments, two Janitors will be hired at 5,000,000 VND each per month. This is a basic salary, as their working hours are limited to specific times of the day —primarily early mornings and late evenings after all kitchens have closed.

Additionally, two Security Personnel will work in alternating shifts to provide 24/7 protection, each receiving 7,000,000 VND monthly. To ensure the living standards of labors, the investor is expected to increase annual real salaries by inflation rate.

2.4. Finance Resources

To launch the project, a total capital of 1,197,534,850 VND is required. Of this amount, 65% will be funded through the investors' own equity, while the remaining 35% will be financed through a collateral-backed loan from Techcombank. The loan carries a fixed interest rate of 6.6% per annum, with a four-year term. Repayment will be made in equal annual installments, covering both principal and interest.

Name	Interest rate	Criteria	Comparable assessment
ACB	6,9% - 12%	Wide range; rate depends on loan type, customer profile, collateral	Potentially high for some borrowers; only competitive if you qualify for lower end.
VPB	7,9% - 8,6%	Typically floating; flexible loan products; quick disbursement	Good for speed and flexibility but costlier in long term.
VCB	0,095	Often fixed for initial period, then floating; very strong bank reputation	stable and reliable, but expensive for this scale of financing.
TCB	0,066	Likely fixed or semi-fixed; standard commercial loan; collateral required	Lowest among peers; competitive offer for 4-year term; suitable for SME

We selected the loan from Techcombank because it offers the lowest interest rate and total repayment cost. While other banks offer interest rates ranging from 6.9% to 12%, Techcombank's fixed rate of 6.6% per annum provides the most cost-effective financing solution. This significantly reduces the financial burden during the initial phase of the project, allowing for better capital allocation and improved financial stability.

Working capital

- Account Receivable: Extended terms allowed (4% of direct revenue); no delays accepted in liquidation year—full settlement required within two payment periods.
- Cash Balance: Maintained at 0.3% of direct revenue for unexpected costs.
- Account Payable: One-time payment for equipment in Year 0; no AP in following years.
- Policy: Delays over two years are penalized per contract.

3. Operational Costs

3.1. Renovation Cost

Today's ever-evolving culinary market and kitchen rental industry make it essential to invest in secure, sanitary, and user-friendly facilities, enabling the company to elevate service quality, attract clientele, and meet legal requirements. Therefore, the costs, budgeting, and comprehensive renovation of the project include the following main items:

- **Electrical and Water Systems:** Over an area of 720 m², the unit price for electrical and water installation is 400,000 VND/m², resulting in a total of 288,000,000 VND
- **Firefighting System:** The total cost for the firefighting system is 11,163,450 VND (not broken down by unit price).
- **Plaster Wall:** The unit price for plaster walls is 175,000 VND/m² for an area of 313.6 m², amounting to 54,880,000 VND.
- **Sem Wall:** The unit price for Sem walls is 350,000 VND/m² for an area of 19.2 m², amounting to 6,720,000 VND.
- **Water Treatment System:** The unit price for water treatment equipment is 4,000,000 VND/m³, with a volume of 5 m³, amounting to 6,720,000 VND.
- **First Aid Kit:** The unit price for each first aid kit is 650,000 VND; the required quantity is 5 kits, amounting to 6,720,000 VND.

In total, when summing all of the above amounts, the renovation cost reaches 299,163,450 VND.

3.2. Management Office & Operating cost

According to estimates, in the first year (Year 0), the total cost for management and sales activities – including office operations, technology equipment, management software, internet service, and stationery – is approximately VND 46 million. This cost is calculated based on current prices and will increase annually with inflation. By the year of liquidation, the projected expense for management and sales will be only 30 percent of the inflation-adjusted cost from the preceding year.

Specifically, the details of the items and costs are as follows:

- **Technology and license costs:** Buy Microsoft 365 package for business (Microsoft for Business) for VND 6,864,000/year. Buy Misa 2025 accounting software (Misa Accounting System) for VND 4,650,000/year. The total for this segment is VND 11,514,000 per year.
- **Internet fee:** Use Viettel's Wifi 2.4/5G package, with a fee of VND 2,640,000/year.
- **Office operating costs (including water, electricity, internal facility usage costs, etc.):** Rent and use an office at Office Saigon with a fee of VND 7,920,000/year in normal usage mode.
- **Marketing costs to boost sales when the warehouse is empty:** Use Haravan's service in the "Small enterprise" package with a cost of 10,000,000 VND/year.
- **Office supplies costs:** Purchase common items (stationary) from many different sources (multi-sources), total cost 24,000,000 VND/year.

Therefore, the total cost for the above items in the first year (Year 0) is VND 56,074,000. In subsequent years, this amount will be adjusted upward according to inflation. By the year of liquidation, the cost of maintaining management and sales activities will be only 30 percent of the inflation-adjusted amount from the previous year.

4. Legal Framework

No.	Law/Regulation Name	Main Content	Practical Application at SHARED KITCHEN 4.0
1	Enterprise Law No. 59/2020/QH14	Business registration, conditional business sectors	Registered as a joint-stock company. Industry codes: 5610 (Restaurants and mobile food services), 5621 (Contracted catering services). Company charter, Appendices I-4 and I-7, founders' list and personal documents completed.
2	Food Safety Law No. 55/2010/QH12	Conditions ensuring food safety at production and business establishments; procedure for food safety certification	Signed food safety commitments with tenants. SHARED KITCHEN acts as representative to apply for food safety certificates for the whole kitchen or individual tenants.
3	Decree 15/2018/ND-CP	Guidance on Food Safety Law implementation; entities required to have food safety certificates	Kitchens with cooking activities must obtain food safety certificates; submit application dossiers per Article 19.
4	Fire Prevention and Fighting Law No. 27/2001/QH10 (amended 2013)	Responsibilities for fire safety at production and business facilities	Design, appraisal, and acceptance of fire protection systems: heat detectors, smoke detectors, fire extinguishers, automatic fire suppression systems. Fire safety certificates issued by district or provincial police authorities.
5	Decree 136/2020/ND-CP	Detailed regulations on fire prevention conditions and permits	Ensure complete appraisal and acceptance dossiers for cooking equipment and cold storage.
6	Environmental Protection Law No. 72/2020/QH14	Environmental impact assessment, waste treatment requirements	Developed and submitted an environmental protection plan. Proper treatment of wastewater, organic waste, and grease in compliance with regulations. Contracted professional waste treatment services long-term.

7	Circular 02/2022/TT-BTNMT	Guidance on waste classification and treatment	Source waste segregation, treatment of emissions and grease with national technical standards.
8	QCVN 12-1:2011/BYT	National technical regulation on food safety conditions for ready-to-eat food processing facilities	Kitchen area designed according to standards: separated zones for preparation, cooking, and storage.
9	TCVN 5603:2008 (HACCP)	Hazard Analysis and Critical Control Point system for food production safety	HACCP procedures applied for controlling food safety in kitchen operations.
10	QCVN 06:2021/BXD	Construction safety standards for buildings and works	Kitchen layout ensures fire escape routes, fire safety compliance, structural safety according to renovation.
11	Construction Law No. 50/2014/QH13 (amended by Law 62/2020/QH14)	Regulations on construction permits and building repairs	SHARED KITCHEN 4.0 renovated an industrial workshop. Complete renovation permit application submitted.
12	Decree 15/2021/ND-CP	Detailed regulations on construction investment project management	Prepared renovation permit dossier: drawings, land use rights certificates, structural safety reports, fire safety, environmental protection. Renovation permit granted on May 25, 2025.

Upon thorough review and comprehensive understanding of the legal requirements and permits necessary for project establishment, the group hereby intends to establish a legal entity in the form of a joint stock company to operationalize the business model, with the shareholders being the aforementioned members. The detailed information is as follows:

Legal Entity Formation Plan

Company Name:

SHARED KITCHEN 4.0 JOINT STOCK COMPANY

Registered Address:

xx Street, Ward xx, Tan Phu District, Ho Chi Minh City, Vietnam

Charter Capital:

VND 5,000,000,000 (in words: Five billion Vietnamese Dong)

Legal Representative:

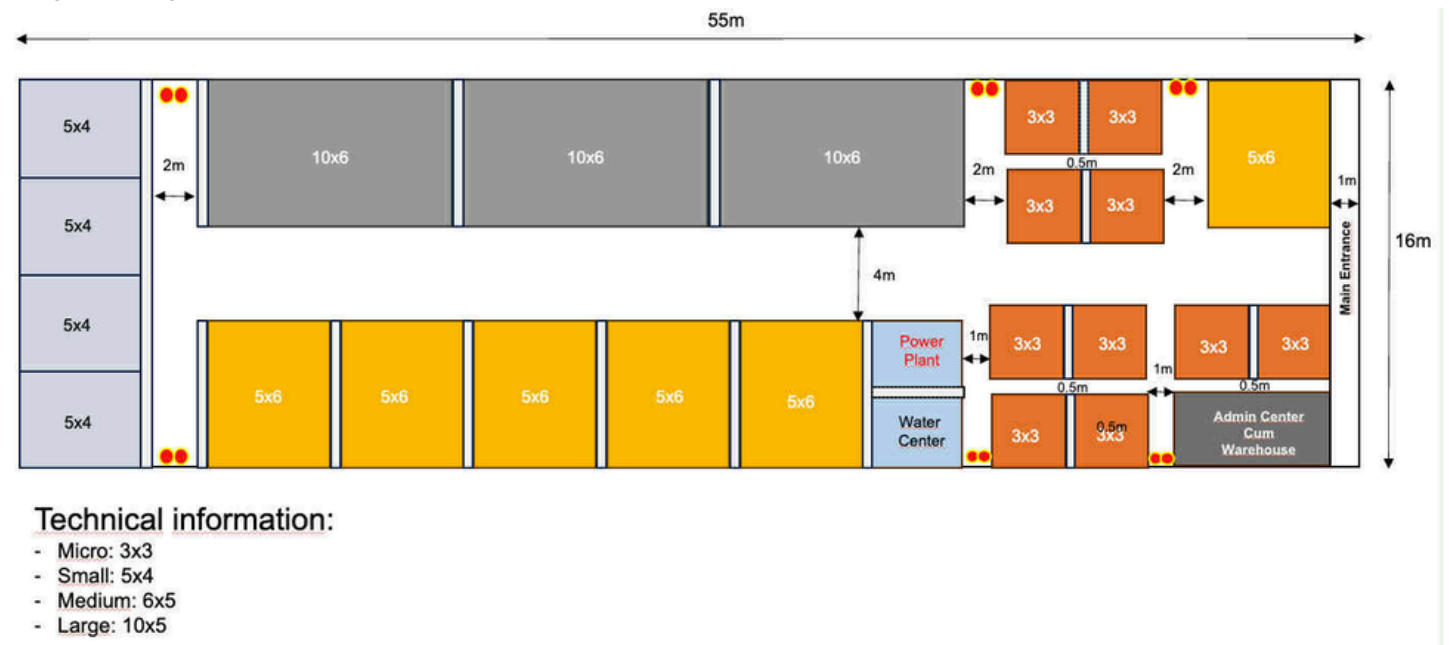
Mr. Huynh Tai Duc

Position: Chairman of the Board of Directors cum General Director

Registered Business Sectors

No.	Business Sector Description	Sector Code	Selected (X)
1	Provision of catering services under non-recurring contracts with clients	5621	X
2	Restaurants and mobile food service activities	5610	
3	Beverage serving services	5630	

Project Layout



The following layout represents the finalized design of the Shared Kitchen 4.0 facility, developed after thorough planning and evaluation. The spatial arrangement and functional zoning have been strategically optimized to maximize operational efficiency and profitability.

5. Rental Service

The core business model of the project is centered around the rental of commercial kitchen spaces, designed to cater to a wide range of customer needs. The primary source of revenue will be derived from kitchen rental fees, which are segmented based on different space sizes and strategic locations within the facility.

Rental pricing is determined by factors such as kitchen size, layout, and placement within the premises, and is structured as follows:

Name of the segment	Code of unit	Quantity	Ground floor area (sqm)	Price (VND/ month)	Detail
Standard large	12-14	3	60	₫25,000,000	The most advanced kitchen with high capacity and massive volume of production per day, placed in the central of the zone, far away from the Waste treatment and highly approachable for shipping agents
Standard medium	15-19	5	30	₫15,000,000	Located in the central of the zone but they don't have the frontage and take time to deliver final product for shipping and have to use cargo to get the ingredients
Standard small	20-23	4	20	₫8,500,000	Located at the leftside of the map and at the corner, take long distance for delivery and far from the entrance and this zone is a compound so it's far the exit door
Special micro	2	1	9	₫6,500,000	Expose the frontage and the most expensive site of the area where it's easy for sales and delivery men to pick up, easy for material fill up and can use the frontage for marketing, but it's smaller than other segments
Executive micro	3	1	9	₫6,000,000	Also a sub-category of micro, but it doesn't expose directly to the frontage so the price of it is lower than the special one in the same segmentation

The implementation roadmap for the Shared Kitchen 4.0 project is strategically structured over a 12-month period from January to December 2025 (Year 0). It is designed to ensure a smooth transition from concept development to full operational capacity, emphasizing regulatory compliance, cost efficiency, and user-focused service delivery.

Phase 1: Planning (January – March 2025)

This phase includes the definition of project objectives, market research, site surveys, and customer segmentation analysis. Key activities are as follows:

- Project Overview & Objectives (1.1 – 1.2): Clarify the project's mission, scope, and operational framework to align the team and future partners.
- Surveying Potential Locations (1.3): This includes visiting sites, assessing infrastructure conditions, and analyzing factors such as foot traffic, delivery radius, and population density.
- Market & Customer Analysis (1.4): Research is conducted to understand the demand for flexible kitchen rentals among food startups, online vendors, and home-based producers. Persona profiles and user pain points are identified to shape the service offering.
- Design Finalization (1.5): Based on user needs and operational goals, the layout and functionalities of the kitchen are finalized, including zoning (cold storage, prep area, cooking line, dishwashing, etc.).
- Financial Allocation (1.6): This includes budgeting, selecting suppliers, negotiating service contracts, and evaluating interest rates for any external financing.
- Business Registration (1.7): Early registration ensures legal clarity for licensing, contracts, and banking.

Phase 2: Construction (March – June 2025)

The second phase focuses on physical and digital infrastructure development.

- Renovation (2.1) begins in March and includes structural adjustments, plumbing, electricity upgrades, ventilation systems, and food-grade flooring and walls.
- Equipment Installation (2.2) begins in May, featuring industrial-grade stoves, cold storage units, stainless-steel prep tables, waste disposal systems, and other critical appliances.
- Licensing & Inspection (2.3): The facility undergoes the process of acquiring a Food Safety Certificate (VSATTP) and fire safety clearance, which are essential for regulatory compliance and platform partnerships (e.g., GrabFood, ShopeeFood).
- Software Implementation (2.4): A lightweight ERP or management software is installed to support functions like kitchen booking, automated billing, hygiene checklists, and energy usage monitoring.

Phase 3: Pilot Testing (July – September 2025)

Pilot testing is essential to validate the shared kitchen model, identify operational weaknesses, and refine the user experience.

- Soft-Launch Marketing (3.1) begins in July, targeting specific user groups such as food delivery vendors, home-based chefs, and small-scale caterers. The campaign includes both digital and offline touchpoints: Facebook ads, in-market booths, and direct outreach to food entrepreneurs.
- User Feedback Collection (3.2) starts in August, using surveys, interviews, and app analytics to gather insights on usability, space layout, equipment performance, and scheduling systems.
- Issue Resolution (3.2.1): Technical issues and process inefficiencies identified during testing are addressed. Adjustments may include rescheduling peak usage hours, updating equipment, or revising SOPs for cleanliness and access control.

Phase 4: Full-Scale Rollout (October – December 2025 - 2035)

Following pilot validation, the model enters full operation in Q4.

- Operational Monitoring (4.1) ensures that kitchen facilities function smoothly with real-time troubleshooting support. Daily maintenance, cleaning schedules, and energy monitoring are closely tracked.
- Data Collection & Analytics (4.2): Usage patterns, tenant behavior, and operational bottlenecks are analyzed to improve efficiency. This data also informs future investment decisions and pricing strategy.
- Customer Support (4.3): A dedicated support hotline is established to assist users with onboarding, equipment training, and technical issues.
- Loyalty Programs (4.4) are launched to retain tenants and encourage longer contract durations. These may include discounts for repeat customers or bundled services like delivery packaging or digital POS setup.
- Performance Reviews (4.5) are conducted in November and December to evaluate financial health, occupancy rates, customer satisfaction, and operational KPIs. These reviews guide strategic decisions for 2026, including potential expansion.

IV. FINANCIAL ANALYSIS

1. Revenue structure

The table below is the project's revenue table, developed based on the rental prices of each kitchen segment and the estimated occupancy rates by year. Rental prices have been adjusted for inflation and include an additional 3% annual increase. The revenue figures reflect actual expectations by multiplying the adjusted rental rates with the projected occupancy rates each year.

Year	0	1	2	3	4	5	6	7	8	9
Standard Large	900,000,000	960,650,100	1,025,387,350	1,094,487,178	1,168,243,574	1,246,970,340	1,331,002,425	1,420,697,347	1,516,436,720	1,618,627,875
Special Medium	216,000,000	230,556,024	246,092,964	262,676,923	280,378,458	299,272,882	319,440,582	340,967,363	363,944,813	388,470,690
Standard Medium	900,000,000	960,650,100	1,025,387,350	1,094,487,178	1,168,243,574	1,246,970,340	1,331,002,425	1,420,697,347	1,516,436,720	1,618,627,875
Standard Small	408,000,000	435,494,712	464,842,265	496,167,521	529,603,754	565,293,221	603,387,766	644,049,464	687,451,313	733,777,970
Special Micro	78,000,000	83,256,342	88,866,904	94,855,555	101,247,776	108,070,763	115,353,543	123,127,103	131,424,516	140,281,082
Executive Micro	72,000,000	76,852,008	82,030,988	87,558,974	93,459,486	99,757,627	106,480,194	113,655,788	121,314,938	129,490,230
Standard Micro	528,000,000	563,581,392	601,560,578	642,099,144	685,369,563	731,555,933	780,854,756	833,475,777	889,642,876	949,595,020
Total Sales		2,152,176,441	2,403,234,511	2,640,632,731	3,221,236,948	3,438,312,885	3,670,017,352	4,651,836,680	4,965,319,302	5,299,927,204

2. Cost structure

As mentioned in the project proposal, the total cost of equipment purchase and installation is a one-time payment in the year of investment and will be evenly distributed over 10 years of operation in the form of depreciation The economic lifetime of these properties is 15 years.

Year	0	1	2	3	4	5	6	7	8	9
Beginning value		748371400	698479973	648588547	598697120	548805693	498914267	449022840	399131413	349239987
Depreciation		49891427	49891427	49891427	49891427	49891427	49891427	49891427	49891427	49891427
Accumulated depreciation		49891427	99782853	149674280	199565707	249457133	299348560	349239987	399131413	449022840
Ending value	748371400	698479973	648588547	598697120	548805693	498914267	449022840	399131413	349239987	299348560

The depreciation schedule outlines the annual depreciation of the equipment over a 10-year period. The initial asset value in Year 0 is about 74.8 billion VND. Under the straight-line depreciation method, the annual depreciation expense for kitchen equipment is 49,891,427 VND. The ending asset value declines correspondingly and is projected to be approximately 29.9 billion VND end of Year 9.

The salary expenses in the table below are calculated on an annual basis, and the wage levels for each employee tier have been adjusted in accordance with the inflation rate.

Year	0	1	2	3	4	5	6	7	8	9
Admin & manager		18,653,400	19,330,518	20,032,216	20,759,386	21,512,951	22,293,872	23,103,139	23,941,783	24,810,870
Maintenance technician and mechanic		20,726,000	21,478,354	22,258,018	23,065,984	23,903,279	24,770,968	25,670,155	26,601,981	27,567,633
Janitor		10,363,000	10,739,177	11,129,009	11,532,992	11,951,640	12,385,484	12,835,077	13,300,991	13,783,817
Securityman		14,508,200	15,034,848	15,580,613	16,146,189	16,732,296	17,339,678	17,969,108	18,621,387	19,297,343
Total labor cost		64,250,600	66,582,897	68,999,856	71,504,551	74,100,166	76,790,002	79,577,479	82,466,141	85,459,662

The consolidated cost structure of the project is presented in the table below. Costs are categorized into operating expenses and management & sales expenses. Although equipment purchases represent a significant expenditure, they have been allocated through depreciation. Therefore, the total cost summary shown below excludes equipment costs, as they are already reflected in the depreciation schedule.

Year	0	1	2	3	4	5	6	7	8	9	10
Land rental costs	450,000,000	951,323,400	1,005,573,568	1,062,917,407	1,123,531,335	1,187,601,832	1,255,326,015	1,326,912,236	1,402,580,733	1,482,564,302	
Labor costs		64,250,600	66,582,897	68,999,856	71,504,551	74,100,166	76,790,002	79,577,479	82,466,141	85,459,662	
Total operating cost	450,000,000	1,015,574,000	1,072,156,465	1,131,917,262	1,195,035,885	1,261,701,998	1,332,116,016	1,406,489,715	1,485,046,874	1,568,023,964	
Management and Sales cost		148,391,942	153,778,570	159,360,732	165,145,526	171,140,309	177,352,702	183,790,605	190,462,204	197,375,982	59,212,795
Total cost	450,000,000	1,163,965,942	1,225,935,035	1,291,277,994	1,360,181,412	1,432,842,307	1,509,468,719	1,590,280,320	1,675,509,079	1,765,399,947	59,212,795

3. Loan schedule

The loan was initially disbursed at the beginning of Year 0, with a beginning balance of 419.14 billion VND (35% of total investment). Interest accrues annually and is fully paid each year, indicating that the loan follows a fixed annual payment plan, including both principal and interest.

Year	0	1	2	3	4
Beginning balance		419,137,198	329,487,122	230,423,789	120,958,806
Accrued interest		44,009,406	34,596,148	24,194,498	12,700,675
Payments		133,659,481	133,659,481	133,659,481	133,659,481
+ Principal repayment		89,650,075	99,063,333	109,464,983	120,958,806
+ Interest payment		44,009,406	34,596,148	24,194,498	12,700,675
Ending balance	419,137,198	329,487,122	230,423,789	120,958,806	0

Using principal+interest payment method, annual payments remain constant at approximately 133.66 billion VND, while the portion allocated to interest decreases over time due to the declining principal balance. In contrast, the principal repayment increases each year, accelerating debt reduction.

4. Cash flow analysis

4.1 Cashflow statement

Year	0	1	2	3	4	5	6	7	8	9	10
CASH INFLOWS (CIFs)											
Total sales	0	2,152,176,441	2,403,234,511	2,640,632,731	3,221,236,948	3,438,312,885	3,670,017,352	4,651,836,680	4,965,319,302	5,299,927,204	0
Salvage value											427,594,544
Rental deposit refund											150,000,000
Change in AR		(86,087,058)	(10,042,323)	(9,495,929)	(23,224,169)	(8,683,037)	(9,268,179)	(39,272,773)	(12,539,305)	(13,384,316)	0
Total cash inflows	0	2,066,089,383	2,393,192,188	2,631,136,802	3,198,012,780	3,429,629,848	3,660,749,173	4,612,563,907	4,952,779,997	5,286,542,888	577,594,544
CASH OUTFLOWS (COFs)											
Renovation and setup	299,163,450										
Machinery and equipments	748,371,400	0									
Rental	450,000,000	951,323,400	1,005,573,568	1,062,917,407	1,123,531,335	1,187,601,832	1,255,326,015	1,326,912,236	1,402,580,733	1,482,564,302	0
Rental deposit	150,000,000										
Total operating cost	450,000,000	1,015,574,000	1,072,156,465	1,131,917,262	1,195,035,885	1,261,701,998	1,332,116,016	1,406,489,715	1,485,046,874	1,568,023,964	0
Management and sales cost	0	148,391,942	153,778,570	159,360,732	165,145,526	171,140,309	177,352,702	183,790,605	190,462,204	197,375,982	59,212,795
Change in CB	0	6,456,529	753,174	712,195	1,741,813	651,228	695,113	2,945,458	940,448	1,003,824	0
Tax	0	98,840,219	228,540,666	265,032,048	369,670,972	401,094,116	432,109,727	612,311,272	657,962,045	706,905,452	0
Total cash outflows	2,097,534,850	2,220,586,090	2,460,802,443	2,619,939,643	2,855,125,531	3,022,189,483	3,197,599,573	3,532,449,286	3,736,992,304	3,955,873,524	59,212,795
NET CASH FLOWS TIPV	(2,097,534,850)	(154,496,707)	(67,610,255)	11,197,159	342,887,248	407,440,365	463,149,600	1,080,114,621	1,215,787,693	1,330,669,365	518,381,749

In Year 0, the project recorded the most substantial negative cash flow at 2,097.5 billion VND, primarily driven by upfront investment costs such as renovation and setup (299 billion VND), machinery and equipment purchases (748 billion VND), rental deposit (150 billion VND), and rental expenses (450 billion VND).

From Year 1 to Year 2, net cash flows remain negative (154 billion VND and 67 billion VND, respectively), mainly due to insufficient cash inflows (CIFs) to cover outflows (COFs), especially rental costs, operating expenses, and taxes.

Starting from Year 3, net cash flows turn positive and increase significantly year by year, indicating that operations have stabilized and the revenue stream has become sufficient to cover expenses. Particularly from Year 5 onward, annual net cash flows consistently exceed 400 billion VND, peaking in Year 9 at about 1,330 billion VND.

Year	0	1	2	3	4	5	6	7	8	9	10
CASH INFLOWS (CIFs)											
Total sales	0	2152176441	2403234511	2640632731	3221236948	3438312885	3670017352	4651836680	4965319302	5299927204	0
Salvage value	0	0	0	0	0	0	0	0	0	0	427594544
Rental deposit refund	0	0	0	0	0	0	0	0	0	0	150000000
Change in AR	0	(86087058)	(10042323)	(9495929)	(23224169)	(8683037)	(9268179)	(39272773)	(12539305)	(13384316)	0
Loan disbursement	419137198										
Total cash inflows	419137198	2066089383	2393192188	2631136802	3198012780	3429629848	3660749173	4612563907	4952779997	5286542888	577594544
CASH OUTFLOWS (COFs)											
Renovation and setup	299163450	0	0	0	0	0	0	0	0	0	0
Machinery and equipments	748371400	0	0	0	0	0	0	0	0	0	0
Rental	450000000	951323400	1005573568	1062917407	1123531335	1187601832	1255326015	1326912236	1402580733	1482564302	0
Rental deposit	150000000	0	0	0	0	0	0	0	0	0	0
Total operating cost	450000000	1015574000	1072156465	1131917262	1195035885	1261701998	1332116016	1406489715	1485046874	1568023964	0
Management and sales cost	0	148391942	153778570	159360732	165145526	171140309	177352702	183790605	190462204	197375982	59212795
Change in CB	0	6456529	753174	712195	1741813	651228	695113	2945458	940448	1003824	0
Tax	0	98840219	228540666	265032048	369670972	401094116	432109727	612311272	657962045	706905452	0
Payment		133659481	133659481	133659481	133659481						
Total cash outflows	2097534850	2354245571	2594461924	2753599124	2988785012	3022189483	3197599573	3532449286	3736992304	3955873524	59212795
NET CASH FLOWS EPV	(1678397653)	(288156188)	(201269736)	(122462322)	209227768	407440365	463149600	1080114621	1215787693	1330669365	518381749

In the investment phase, the primary cash inflow came from a loan disbursement of approximately 41.9 billion VND. However, the total initial investment cost was substantial, including renovation and setup expenses (around 30 billion VND), machinery and equipment (over 74.8 billion VND), along with rental fees and deposit payments. As a result, total cash outflows in year 0 reached VND 209.7 billion, leading to a net cash flow deficit of VND -167.8 billion VND, marking a highly favorable financial conclusion to the project's life cycle.

In the following two years (years 1 and 2), the project began generating rental income, yet net cash flow remained negative at 288.1 billion and 201 billion, respectively. This reflects a phase in which the project was still scaling up operations, completing its system setup with the occupancy rate at about 65% and 68%, respectively, and had yet to reach a breakeven revenue level.

From year 3 to year 9, the project enters a stable and growth-driven phase with consistently positive net cash flows. Starting with 20.9 billion VND in year 3, cash flows steadily rise to over 133 billion VND by year 9. This period reflects strong operational efficiency, increasing occupancy rates, and optimized cost management, marking the project's most profitable and mature stage.

In the liquidation year, cash inflows surge due to two significant sources: the refund of rental deposits (15 billion) and the asset salvage value at about 427.6 billion. As a result, net cash flow peaked at about 518.4 billion VND, marking a highly favorable financial conclusion to the project's life cycle.

4.2 Project assessment criteria

The net present value (NPV), under both TIPV and EPV approaches, stands at 63.2 billion VND and 61.3 billion VND, respectively. Both values are significantly positive, indicating that the project generates considerable added value over the initial investment, confirming its profitability and economic efficiency.

The benefit-cost ratio (BCR) is calculated under 2 views:

- View 1 calculates BCR by dividing the NPV of total cash inflows by the NPV of total cash outflows.

- View 2 takes a more investment-focused perspective by dividing the NPV of operational cash flows by the NPV of investment cash flows.

In both approaches, the BCR values exceed 1, confirming that the project's benefits outweigh its costs and making it a financially sound investment.

Net present value		Benefit-cost ratio (TIPV)	
NPV (TIPV)	632485857	BCR (View 1)	1.03
NPV (EPV)	613424046	BCR (View 2)	1.30

IRR	
IRR (TIPV)	12.20%
IRR (EPV)	12.30%

Using the non-discounted method to calculate the payback period, the Shared-Kitchen 4.0 project demonstrates a payback period of 3.14 years under the view of TIPV and 3.79 years under the view of EPV. This means the project is expected to recover its initial investment in just over 3 years (approximately 3 years, 1 month, and 19 days for TIPV and 3 years, 9 months, and 13 days for EPV).

These periods indicate that the project's cash flow generation is robust and timely enough to service its financial obligations without significant delays or cash strain. Overall, the results confirm the project's strong potential in terms of both investment recovery and debt servicing capacity.

Year	0	1	2	3	4	5	6	7	8	9	10
Accumulated NCF (TIPV)	(2,097,534,850)	(2,252,031,557)	(222,106,962)	(56,413,096)	354,084,407	750,327,613	870,589,965	1,543,264,221	2,295,902,314	2,546,457,057	1,849,051,114
Accumulated NCF (EPV)	(1,678,397,653)	(1,966,553,840)	(489,425,924)	(323,732,058)	86,765,445	616,668,132	870,589,965	1,543,264,221	2,295,902,314	2,546,457,057	1,849,051,114
PP (TIPV)		3.14									
PP (EPV)		3.79									

TIPV	Years	Months	Days
	3	1	19
EPV	Years	Months	Days
	3	9	13

Annual debt service coverage ratio (ADSCR) TIPV

Year	0	1	2	3	4
ADSCR		-1.72	-0.68	0.10	2.83

The Annual Debt Service Coverage Ratio (ADSCR) measures the project's ability to generate sufficient cash flow to cover its annual debt obligations. According to the table, in the first two years (Year 1 and Year 2), the ADSCR values are -1.72 and -0.68, respectively. These negative figures indicate that the project's operating cash flows are insufficient to meet its debt service requirements, implying a reliance on additional funding or reserves during the initial phase.

From Year 3, the ratio turns positive (0.10) and significantly improves to 2.83 by Year 4, exceeding the commonly accepted safety threshold of 1.0. This suggests that after the initial investment and start-up phase, the project begins to generate stable cash flows that are sufficient not only to cover debt payments but also to provide surplus for reinvestment or savings.

V. ANALYZE PROJECT RISKS

1. Introduction

Risk assessment constitutes a fundamental project evaluation component, particularly in contexts characterized by uncertainty and variability in key input parameters. In fact, our project is established and formed in a new business model in Vietnam, so it could contain a lot of underlying and potential threats that could affect the profitability and feasibility of this model. Traditionally, in Vietnam, we already have the same model in party planning agents with a smaller size and less initial cost, but it runs spontaneously and lasts for a short duration. This session aims to systematically investigate the sensitivity of project outcomes to changes in critical variables by applying quantitative risk analysis techniques. Specifically, one-way and two-way sensitivity analyses were employed to identify and compare the influence of individual and paired variables on the project's performance metrics. Scenario analysis was conducted to evaluate the implications of alternative future states, including best-case, worst-case, and most-likely conditions. Furthermore, a Monte Carlo simulation, implemented via Oracle Crystal Ball, was utilized to model probabilistic outcomes by incorporating uncertainty into multiple input variables simultaneously. The integration of these methods provides a comprehensive framework for understanding the range and likelihood of potential project outcomes, thereby supporting more robust decision-making and strategic planning.

2. Detailed methodology

2.1. One-Way Sensitivity Analysis

The one-way sensitivity analysis was conducted in this project to evaluate the impact of some potentially risky variables that could have a significant impact on the Net Present Value (NPV), which is the most important and common criterion to evaluate whether this project could attract investors and call for capitalization or not. In this project, the revenue structure is quite simple since we just get our revenue as a management service supplier where tenants rent some sites in our shared kitchen zones, so the main variables that we could easily identify are: price of the kitchen's segmentations, some main cost such as labor cost, management and sales cost, and more important is the occupancy rate in the first year of operation. The reason why the occupancy rate in year 1 is essential in the shared kitchen model is that: in most successful countries that have already run this type of business, such as Singapore, America... report that the occupancy rate is the main index showing how the business is accepted, the first year of operation is the most valuable time to understand the market performance and if the occupancy rate in the first year is high, it leads to the high occupancy rate follows in the next year and vice versa. And in our one-way analysis, we do not just care about the NPV, we take into consideration some other ones including BCR, IRR as well to have the overview about how each variable changes affecting the whole project.

By applying one-way sensitivity analysis for the above listed variables, the results illustrate that:

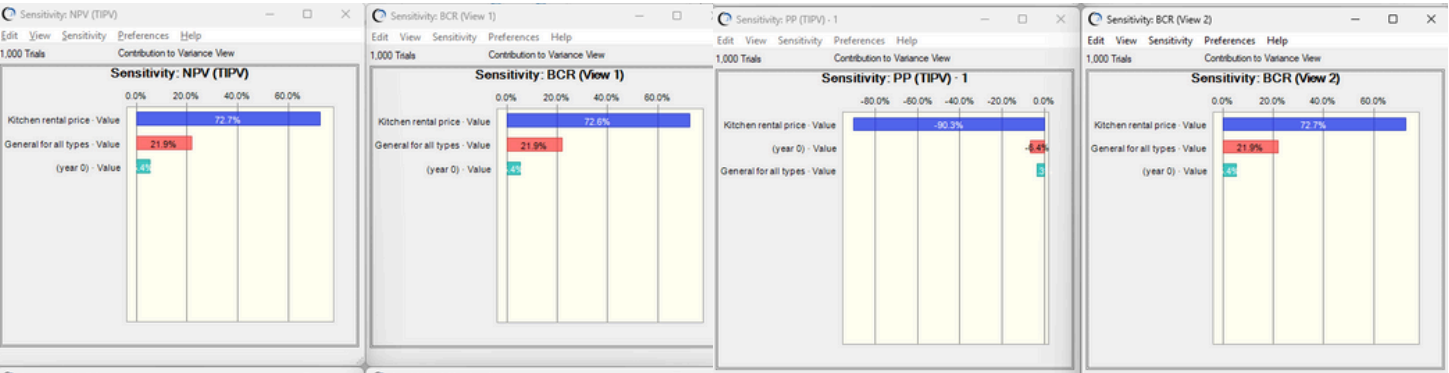
Variables	Sensitivity result	Assessment
Price of Large kitchen	- When the price of this segment changes, NPV changes with bigger amount, for example: when the price increases by 6%, NPV (TIPV) increases by 45,2% - This trend is true for BCR, IRR as well	Since this is the most important segment of kitchen with the highest price charged per tenants, so it's easy to understand that just a little fluctuation in the price that will greatly affect the whole project
Price of Special Medium kitchen	When the price of the Special Medium Kitchen changes, the NPV also changes significantly. For example, when the price increases by 11%, the NPV (TIPV) increases by 20.1%, and when the price decreases by 11%, the NPV (TIPV) drops by -30.1%. This trend is consistent across other indicators such as BCR and IRR. As shown in the table, small price fluctuations lead to substantial impacts on the investment efficiency indicators.	Given that this is the segment with a relatively high price charged per tenant, it's understandable that even minor pricing adjustments can considerably influence the financial performance of the entire project.
Price of Standard Medium kitchen	The Standard Medium Kitchen segment shows high sensitivity to price changes, with a 20% increase in price leading to a 150.7% rise in NPV (TIPV) and strong gains in BCR and IRR. Conversely, a 20% price drop results in a 150.5% decrease in NPV, with BCR falling below 1.0 and IRR turning negative.	As a key revenue-generating segment, even small price adjustments here have a significant impact on project feasibility. Pricing strategy in this segment is therefore critical to the overall success of the investment.
Interest rate	The project demonstrates moderate sensitivity to interest rate fluctuations. A decrease from 10.5% to 5% results in a 62.9% increase in NPV (TIPV) and notable improvements in both BCR and IRR. However, even at higher interest rates, NPV remains positive and BCR > 1, indicating the project's resilience and financial viability.	Interest rate changes affect financing costs, but the project's cash flows maintain strong economic performance within the tested range.
Salary of Manager	Changes in the manager's salary have a minimal impact on the project's financial performance. Even with a 122% increase, the NPV (TIPV) decreases by only ~20%, and BCR remains above 1.025. All indicators (NPV, BCR, IRR) show limited sensitivity, proving that manager salary, though the highest among employees, does not significantly influence overall project viability.	This suggests that the project can absorb changes in high-end labor costs without major financial risk.
Management and sales cost	Although the manager's salary is the highest individual salary, adjustments to it result in minimal changes to financial indicators such as NPV and IRR. On the other hand, management and sales costs have a stronger impact. An increase of 11.74% in these costs causes a 15.8% decrease in NPV (TIPV) and a 4.8% reduction in IRR (TIPV), showing higher sensitivity.	Project profitability is not significantly affected by the manager's salary but is highly sensitive to changes in management and sales costs. Therefore, it is crucial to control these costs to ensure financial sustainability.
Occupancy rate in year 1	Occupancy rate in year 1 has a significant impact on the project's financial outcomes. When the rate drops from 65% to 40%, NPV (TIPV) plunges by 96.1%, and IRR (TIPV) decreases by 29.6%. Conversely, increasing the occupancy rate to 80% results in a 57.7% rise in NPV (TIPV) and a 21.4% increase in IRR (TIPV). BCR (View2) improves from 1.301 to 1.4755, showing that even a small percentage change in occupancy can heavily influence the project's feasibility.	The occupancy rate in year 1 is one of the most sensitive and influential factors in the project's success. Maintaining a high occupancy rate is crucial, as small drops lead to steep declines in profitability and investment returns. Therefore, strategic planning and aggressive marketing are essential to secure occupancy early in the project's lifecycle.

The one-way sensitivity analysis reveals that occupancy rate in year 1 and kitchen pricing strategies, especially for the Standard Medium, Large kitchen segments and Management and sales, are the most critical factors impacting the project's financial viability. Among all variables tested, occupancy rate in the first year shows the strongest influence, with even small changes causing drastic shifts in NPV and IRR—highlighting its role as a key performance indicator and predictor of long-term success. Pricing changes in the revenue-generating kitchen segments also have a significant impact, emphasizing the need for carefully optimized pricing strategies. In contrast, costs such as manager salary have minimal influence, while interest rate and management & sales costs show moderate sensitivity. These insights underscore that maximizing initial occupancy and optimizing pricing for high-value kitchen segments are essential levers for ensuring project profitability and attracting investors.

2.2. Two-Way Sensitivity Analysis

A two-dimensional sensitivity analysis was conducted to investigate how simultaneous fluctuations in standard large kitchen unit price and management & sales (M&S) costs impact the project's Net Present Value (NPV). The one-way analysis had already identified these two variables as the most critical, exhibiting disproportionate impacts on financial viability. Therefore, a cross-tabulated risk matrix was constructed to evaluate their interaction and combined risk magnitude.

M&S → Kitchen ↓	143.2M	150M	160M	170M	180M	190M	200M
25M	63,248,557	21,862,429	-35,273,805	-92,410,039	-149,546,273	-206,682,507	-263,818,741
26M	23,184,623	-18,201,505	-75,337,739	-132,473,973	-189,610,207	-246,746,441	-303,882,675
27M	-16,879,310	-58,265,438	-115,401,672	-172,537,906	-229,674,140	-286,810,374	-343,946,608
28M	-56,943,243	-98,329,371	-155,465,605	-212,601,839	-269,738,073	-326,874,307	-384,010,541
29M	-97,007,176	-138,393,304	-195,529,538	-252,665,772	-309,802,006	-366,938,240	-424,074,474
30M	-137,071,109	-178,457,237	-235,593,471	-292,729,705	-349,865,939	-407,002,173	-464,138,407



Two-way sensitivity analysis result table

The base-case scenario indicates a large kitchen price of 25,000,000 VND and a management and sales cost of 143,194,000 VND, yielding a positive NPV of 63,248,557 VND

As we vary both variables across their plausible operational ranges, the analysis reveals a highly nonlinear response of NPV. For example, a modest increase in the kitchen price to 26,000,000 VND combined with a slight increase in management and sales cost to 166,000,000 VND leads to a significantly reduced NPV of just 20,889,985 VND. When both values continue to increase to 27,000,000 VND and 180,000,000 VND respectively, the NPV turns negative, reaching -3,133,629 VND. In the worst-case scenario of a kitchen price of 30,000,000 VND and a management cost of 200,000,000 VND, the project generates a large negative NPV of -99,860,231 VND, highlighting the high downside risk.

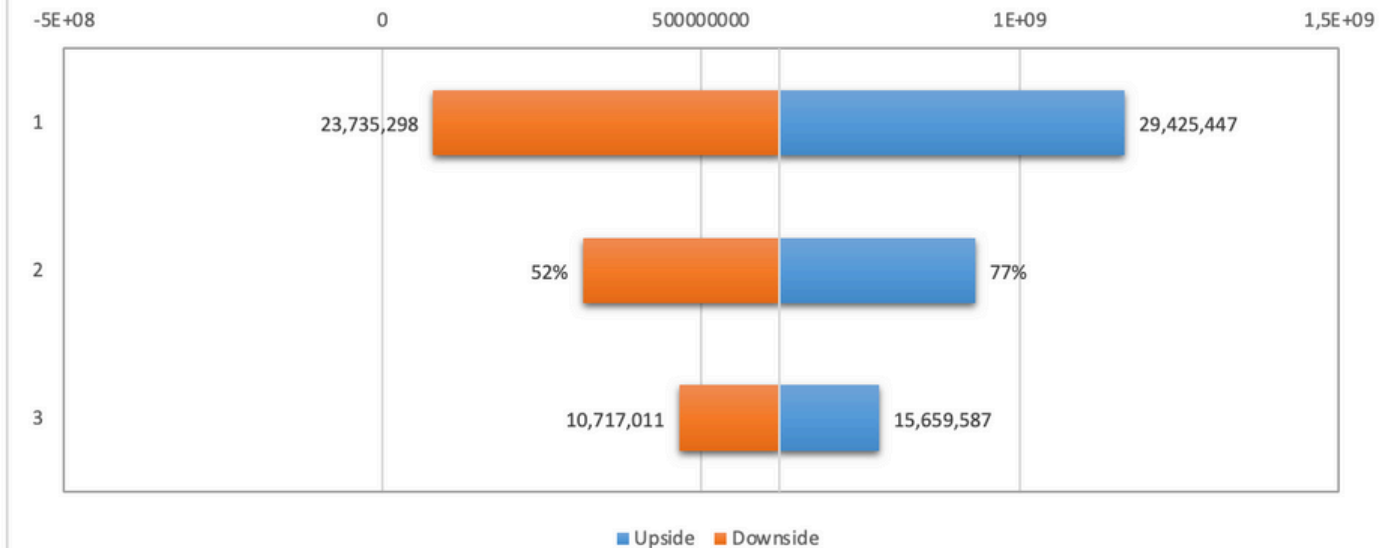
A distinct threshold region is visible around the kitchen price of 25,500,000–26,500,000 VND and management/sales cost of 172,000,000–182,000,000 VND, where the NPV rapidly transitions from positive to negative. For instance, at 25,500,000 VND and 172,000,000 VND, the NPV remains barely positive at 3,084,002 VND. However, increasing the kitchen price to 26,000,000 VND while keeping the same cost level results in an NPV of -14,960,727 VND. The NPV slope along the horizontal axis is steeper, confirming that the project is more sensitive to changes in kitchen price than to variations in management and sales costs. Conversely, under favorable conditions—such as a kitchen price of 23,000,000 VND and overhead of 148,000,000 VND—the NPV reaches 222,365,838 VND, confirming the high upside potential when both cost components are efficiently managed.

This analysis reveals that while each variable individually exerts significant influence on project value, their combined effects can amplify the financial risk. The matrix surface indicates limited resilience zones: only scenarios where both the kitchen price remains below 24,000,000 VND and management costs are contained under 160,000,000 VND exhibit strong and stable positive NPVs. The red-shaded region—representing negative NPVs—spreads diagonally from lower right to upper left, clearly illustrating the tipping point where the joint cost burden becomes unsustainable. More critically, the analysis shows that attempting to compensate for high overhead by reducing unit costs (or vice versa) offers limited mitigation. For example, at a low kitchen price of 23,000,000 VND but high overhead of 192,000,000 VND, the NPV remains negative at -6,153,505 VND, confirming that operational savings cannot independently offset excessive overhead.

The financial feasibility of the project is therefore highly contingent on strict cost control in both capital expenditure and operating expense categories. Without effective procurement strategies and streamlined overhead structures, the risk of value erosion becomes considerable. These results strongly suggest the necessity of proactive risk mitigation: contractual locking of unit prices, performance-based management compensation structures, and lean operational frameworks. Additionally, sensitivity boundaries identified in this analysis should guide the construction of contingency budgets and be reflected in stakeholder risk communications. In summary, the two-way sensitivity analysis underscores a narrow feasibility window, where the project remains viable only if disciplined cost containment is achieved across both parameters.

To further understand how each input variable influenced output variability, sensitivity analysis was conducted using tornado and spider charts. The tornado chart revealed that the kitchen rental price was the dominant risk driver, accounting for 71.12% of the total explained variation in NPV_TIPV. The general price for all kitchen types added further variance, and when combined, these two inputs explained over 94% of the output variation. The occupancy rate in year 0 played a smaller yet still relevant role in overall sensitivity. These findings highlight that pricing strategies, particularly for standard large kitchens, should be actively managed as a key lever of financial performance.

NPV (TIPV)



Input Variable	NPV (TIPV)				Input		
	Downside	Upside	Range	Explained Variation ¹	Downside	Upside	Base Case
Kitchen rental price · Value	79471998	1163944569	1084472571	71,12%	23.735.298	29.425.447	26.580.372
General for all types · Value	313884266	929532111	615647845	94,04%	52%	77%	65%
(year 0) · Value	464709133	778707244	313998110	100,00%	10.717.011	15.659.587	13.188.299

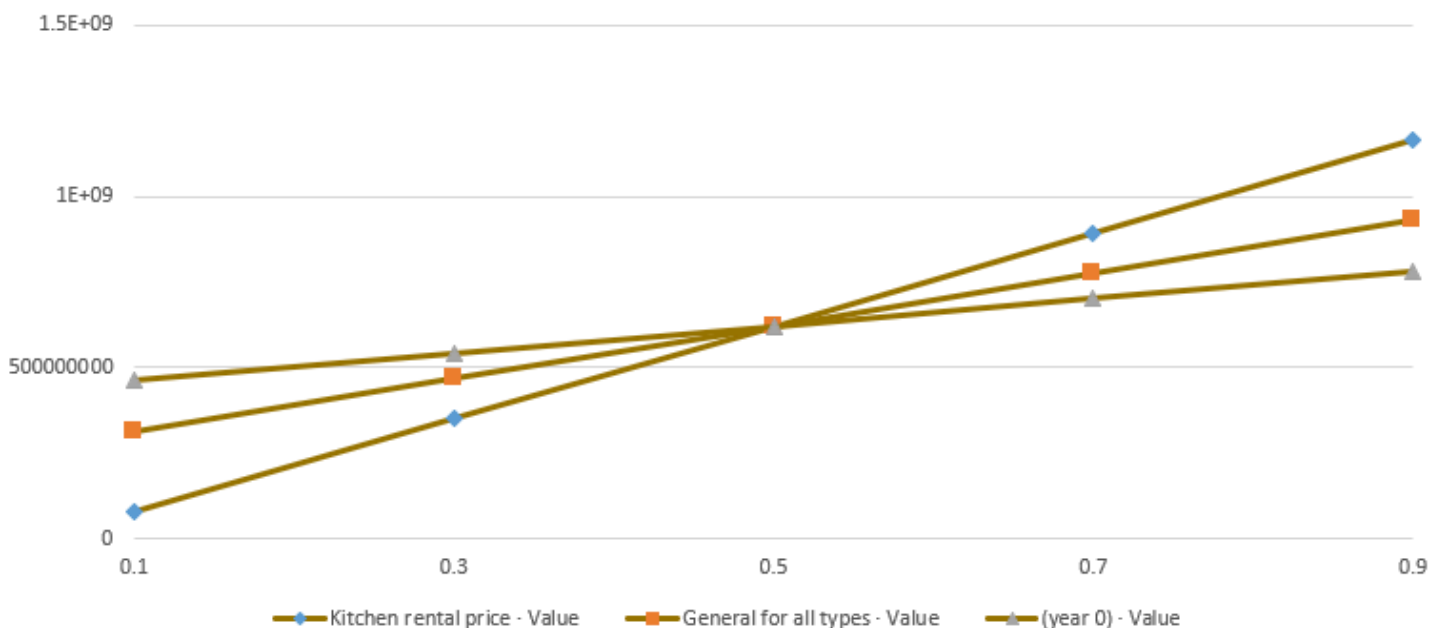
¹ Explained Variation is cumulative

Run options:

Tornado method	Percentiles of the variables
Test range	10% to 90%
Test points	5
Customize test ranges by variable	Off
Show top variables	20
Base case for Crystal Ball variables	Median values

Tornado chart

NPV (TIPV)



Input Variable	Elasticity ¹	NPV (TIPV)				
		10.00%	30.00%	50.00%	70.00%	90.00%
Kitchen rental price · Value	11.00	79471998	350590093	621708189	892826474	1163944569
General for all types · Value	2.67	313884266	467796227	621708189	775620150	929532111
(year 0) · Value	1.35	464709133	543208661	621708189	700207716	778707244

¹Elasticity is averaged across the entire test range

Run options:

Tornado method	Percentiles of the variables
Test range	10% to 90%
Test points	5
Customize test ranges by variable	Off
Show top variables	20
Base case for Crystal Ball variables	Median values

Spider chart

The spider chart reinforced these results by measuring elasticity across different percentiles. The kitchen rental price had the highest elasticity at 11.00, demonstrating that even modest changes in this variable can produce disproportionately large shifts in NPV. The general pricing variable followed with elasticity of 2.67, and the occupancy rate posted a much lower elasticity of 1.35. This elasticity distribution confirms that revenue-side variables dominate the risk landscape and should be prioritized in risk mitigation efforts.

2.3. Scenario Analysis

After conducting the one-way and two-way sensitivity analyses, the next logical step is to explore how the project behaves when multiple critical variables change simultaneously, as this mirrors real-world market complexity more realistically. For this purpose, I created three comprehensive scenarios—Best Case, Common Case, and Worst Case—each representing a possible future state of the project influenced by fluctuations in the occupancy rate in year one, interest rate, land rent, and account receivables. These variables were chosen not arbitrarily, but strategically. The occupancy rate is a direct reflection of market demand and project appeal, and in the first year it is particularly vital since it sets the financial tone for subsequent periods. The interest rate affects both the cost of capital and the project's internal return, which is crucial given the long-term horizon of this investment. Land rent represents a substantial fixed cost burden that cannot be avoided, and any escalation in this cost erodes net profitability. Finally, account receivables play an important role in revenue realization. If tenants do not pay on time, even high occupancy cannot guarantee liquidity or profitability.

Scenario Summary		Current Values:	Best Scenario	Common Scenario	Worst Scenario
			The best case is describe as when AR is smallen down tp 2%, Interest rate decreases by 9.5%, Occupancy rate in year 1 is 80% and Land rent is 70M VND	This is the common case in the marekt where occupancy rate is not too high but not to low, Interest rate is about 12.5%, land rent in general is 80,mVND while AR is about 5% since tenants can not pay all	Worst Scenario is expected to happen when sudden situation appears such as Covid pandemic. In this case, the interest rate is higher at about 15%, occupancy rate in year 1 is just 50%, land rent increased to 85mVND, AR is 8% where tenant can't pay
Changing Cells:					
	\$AC\$15	65%	80%	75%	50%
	\$AC\$20	10,5%	9,5%	12,5%	15,0%
	\$W\$10	75.000.000	70.000.000	80.000.000	85.000.000
	\$W\$34	4%	2%	5%	8%
Result Cells:					
	NPV (TIPV) \$V\$154	632485857	2445923977	-492879069	-1989973755
	NPV (EPV) \$V\$155	613424046	1134809756	-307675874	-1614826121
	BCR (View 1) \$V\$159	1,03	1,13	0,99	0,91
	BCR (View 2) \$V\$160	1,30	2,41	1,17	0,12
	IRR (TIPV) \$V\$164	12,03%	23,00%	8,41%	-0,54%
	IRR (EPV) \$V\$165	12,46%	14,15%	7,72%	-2,09%
	PP (TIPV) \$V\$172	2,82	1,75	4,09	4,57
	PP (EPV) \$V\$173	3,84	3,17	4,12	5,03

Scenario analysis result tables

Key findings:

- Best Case:** Under highly favorable conditions—80% occupancy in year 1, interest rate reduced to 9.5%, and land rent at 70 million VND/month—the project achieves outstanding financial performance. NPV (TIPV) reaches 244.5 billion VND, and NPV (EPV) stands at 113.5 billion VND. IRR (TIPV) is 23.00%, and IRR (EPV) is 14.15%, indicating strong returns on investment. The payback periods are relatively short (2.82 years for TIPV; 3.17 years for EPV), while BCR exceeds 2.0 in both views. This scenario highlights that, if optimal market conditions and effective cost management are sustained, the project offers highly attractive profitability and investment potential.
- Common Case:** In a typical market situation with 75% occupancy, a 12.5% interest rate, and 80 million VND/month land rent, project viability becomes more conditional. NPV (TIPV) turns negative at -49.3 billion VND, and NPV (EPV) also dips to -30.8 billion VND. IRR (TIPV) is modest at 8.41%, while IRR (EPV) drops to 7.72%, suggesting marginal returns. Although BCR (EPV) remains above 1.0 at 1.197, payback periods extend to 4.12 years (TIPV) and 5.03 years (EPV). This scenario implies the project may still be feasible but would likely require financial restructuring, cost optimization, or government/investor support to be attractive.
- Worst Case:** Under stress conditions—including just 50% occupancy in year 1, a high interest rate of 15%, and land rent at 85 million VND/month—the project becomes financially unsustainable. NPV (TIPV) plummets to -198.9 billion VND, and NPV (EPV) reaches -161.5 billion VND. IRR (TIPV) is negative (-0.54%), and IRR (EPV) is barely positive at 1.29%. All BCR values drop below 1.0, and payback periods exceed 4.5 years, signaling critical risks to liquidity and long-term viability. In such conditions, without major strategic pivots or external intervention, the project is unlikely to proceed.

The scenario analysis presents a clear picture of how the shared kitchen project's financial viability responds under varying market conditions. In the Best Scenario, where the occupancy rate reaches 80%, interest rates drop to 9.5%, and land rent decreases to 70 million VND, the project performs exceptionally well, generating an NPV (TIPV) of 244.5

billion VND, BCR (View2) of 2.41, and IRR (TIPV) of 23%, indicating strong profitability and investment attractiveness. This scenario highlights the project's potential under favorable market acceptance and cost conditions. In the Common Scenario, representing a moderate yet realistic outlook (occupancy at 75%, interest at 12.5%, land rent at 80 million VND), the project still remains viable, with a slightly positive IRR (TIPV) of 8.41% and NPV (TIPV) around -49.3 billion VND, although the negative NPV suggests marginal feasibility and signals the need for close cost and revenue control. However, under the Worst Scenario, where interest rates surge to 15%, occupancy plummets to 50%, and land rent rises to 85 million VND, the project becomes financially unsustainable with a severely negative NPV (TIPV) of -198.9 billion VND and IRR (TIPV) turning negative (-0.54%), indicating a clear loss-making venture. These findings confirm that the occupancy rate and interest rate are the two most sensitive and decisive factors in this project, and proactive strategies—such as targeted marketing to boost early occupancy and securing favorable financing—will be essential to mitigate downside risks and maintain investment appeal.

Overall, this scenario analysis demonstrates the importance of preparing for uncertainty with a strong risk mitigation framework. The project shows potential under moderate to optimistic scenarios but faces significant challenges in crisis conditions. The use of multiple scenarios offers not only numerical insights but also a strategic lens through which the investment decision can be evaluated more robustly. The clarity of variation across metrics - NPV, IRR, BCR, and Payback - under the structured assumptions makes it easier to communicate financial viability and risks to investors, stakeholders, and funding partners in a credible and transparent way.

2.4. Monte Carlo Simulation (Crystal Ball)

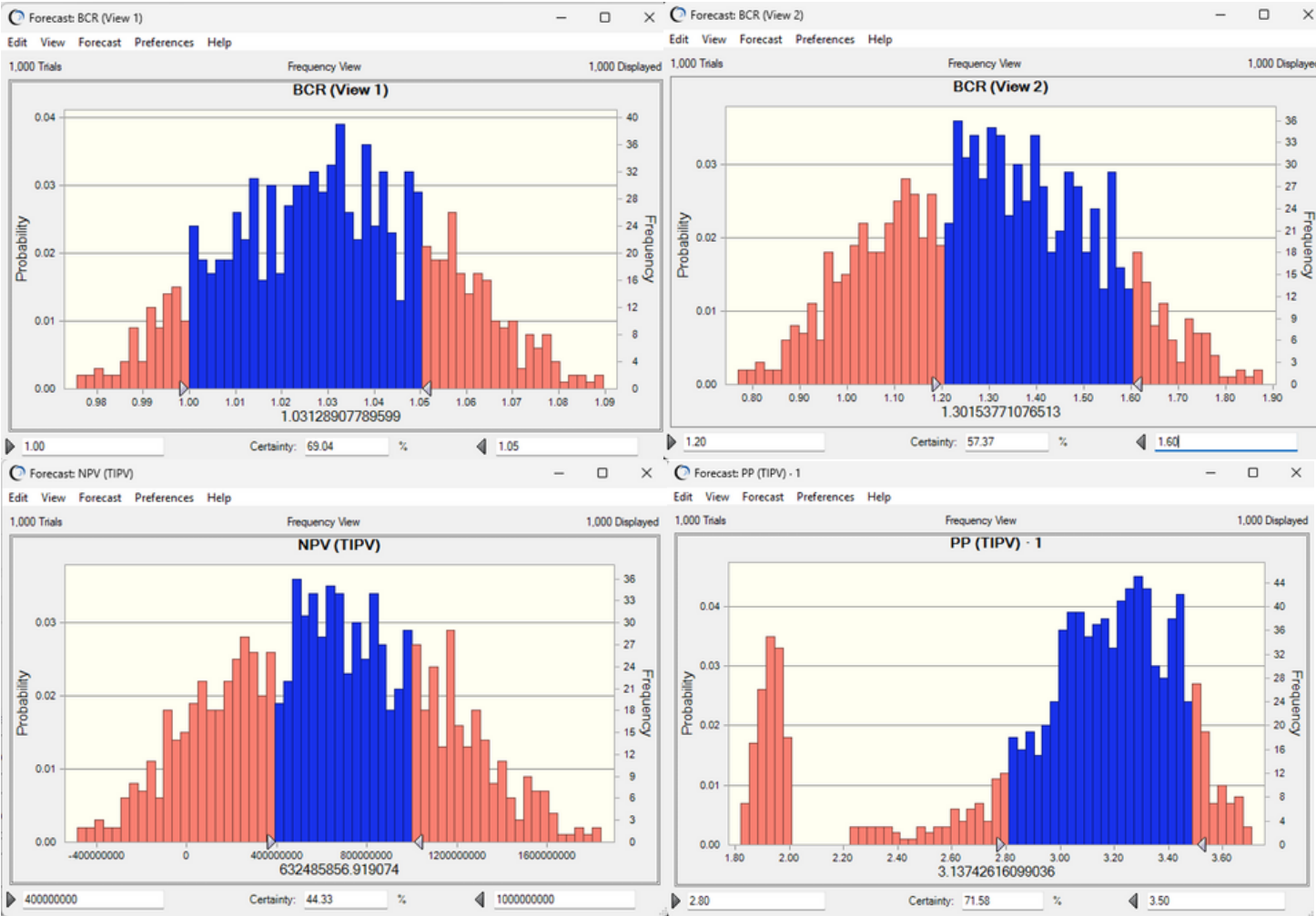
This section presents a risk-adjusted financial evaluation of the investment project using Monte Carlo simulation techniques. The analysis uses the Oracle Crystal Ball to assess the impact of uncertainty on key financial performance indicators, including Net Present Value (NPV), Benefit-Cost Ratio (BCR), and Payback Period (PP). Rather than relying on deterministic projections, this approach integrates probabilistic modeling to simulate a broad range of outcomes under real-world volatility. Historical data was first collected and fitted to appropriate probability distributions to reflect the behavior of critical input variables. The selected inputs include standard large kitchen rental prices, special medium kitchen prices, and initial occupancy rate in year 1 - variables which collectively represent a significant portion of total project revenue and early-stage financial viability. These fitted inputs were standardized before being deployed in simulation, ensuring statistical consistency and representativeness in forecast modeling.

Input variables were chosen based on both their financial materiality and historical volatility. Kitchen rental price was prioritized given its high sensitivity to location and service tier, and because it directly influences projected revenues across all forecast years. The general price for all unit types was similarly included to capture market diversity and price range fluctuation across offerings. Meanwhile, the occupancy rate in year 1 was integrated as an early revenue driver, influencing both cash inflows and cost recovery speed. Each of these variables demonstrated non-trivial distribution spreads, making them suitable for probabilistic modeling. The selection process was guided by the magnitude of their cumulative impact on variance observed in previous deterministic sensitivity tests.

The simulation was conducted with 1,000 trials, and the financial forecasts for each scenario were aggregated and analyzed. The distribution-based outputs for each performance metric are summarized below, focusing on the average result, the simulation certainty, and the full range of possible values produced by the Monte Carlo runs.

Metric	Mean Output	Certainty (%)	Simulated Range (min – max)	Interpretation
NPV_TIPV	632,485,857 VND	4.433%	-400,000,000 to 1,600,000,000 VND	Moderate chance of strong financial gain
BCR (View 1)	1.031	6.904%	0.98 to 1.09	Slightly above breakeven in most cases
BCR (View 2)	1.301	5.737%	0.80 to 1.90	Higher efficiency under alternate assumptions
Payback Period (PP_TIPV)	3.14 years	7.158%	1.80 to 3.60 years	Acceptable recovery timeframe in most scenarios

Monte Carlo Simulation Result table



These figures indicate that, while the base case scenarios offer reasonable returns, there is meaningful variance in the outcomes, especially for NPV. The BCR results suggest the investment is marginally efficient, though improvement in price controls could strengthen the case. The payback period supports a relatively short investment horizon in the majority of outcomes.

In conclusion, the project demonstrates an acceptable risk-reward profile with moderate potential for exceeding baseline financial goals. However, the dependency on a few key price-sensitive variables introduces notable volatility. To increase robustness, strategic attention should be given to securing long-term rental contracts, optimizing pricing segmentation across unit types, and driving occupancy early in the project lifecycle. These strategies will materially reduce downside risk and enhance the project's overall investment quality under uncertainty.

VI. PROJECT'S MULTIDIMENSIONAL IMPACT

1. Economic impact

The Shared Kitchen 4.0 project represents a high-potential economic intervention within the growing F&B (Food and Beverage) industry in Ho Chi Minh City. The financial metrics—particularly the positive Net Present Value (NPV) and Internal Rate of Return (IRR) exceeding 12%—validate the project's profitability under conservative and moderate market scenarios. From an investment evaluation perspective, an IRR above the typical market cost of capital indicates that the project generates a return higher than the minimum acceptable hurdle rate, confirming its attractiveness to both private investors and venture capital. This is especially compelling in Vietnam's current economic context, where inflation is relatively stable, and interest rates are declining to stimulate domestic demand.

In addition to favorable investment indicators, Shared Kitchen 4.0 supports economic decentralization by reducing entry barriers for F&B micro-entrepreneurs and startups. The project utilizes a multi-tenant infrastructure model, which distributes fixed costs across multiple users—effectively increasing capital efficiency and asset turnover. Unlike traditional brick-and-mortar restaurant setups that demand high initial capital and operational expenses, this model provides access to essential kitchen utilities, storage, and logistics for a fraction of the cost. In the long term, this promotes the localization of food production, especially around high-density residential complexes like Saigon City and near industrial zones, including Linh Trung Export Processing Zone and Tan Binh Industrial Park. This proximity reduces distribution lead times and supports the formation of micro food clusters, enhancing regional supply chain resilience and labor productivity.

2. Environmental impact

The social implications of Shared Kitchen 4.0 are significant, particularly in the context of urban consumer behavior and food safety demands in Ho Chi Minh City. With a population exceeding 9 million and rapid suburban expansion, there is an escalating demand for clean, safe, and convenient food services within residential and industrial areas. The kitchen zones are strategically located in close proximity to high-density zones such as Saigon City residential complex and major retail centers like Aeon Mall Tan Phu

Celadon. These zones not only allow rapid food delivery services through on-demand logistics platforms (e.g., Baemin, GrabFood), but also serve as centralized nodes for fresh meal preparation under monitored hygiene conditions.

Moreover, the project contributes to public health and food hygiene improvements through enforced food safety standards and cross-kitchen compliance monitoring. Shared Kitchen 4.0 implements HACCP (Hazard Analysis and Critical Control Points) protocols and offers tenants access to a sterilization and audit system that ensures minimal risk of food contamination. This is especially relevant given Vietnam's frequent food safety scandals and growing consumer awareness. Beyond public health, the project fosters inclusive socio-economic mobility by enabling small food operators—many of whom may be informal workers, women, or family-run businesses—to scale their operations without the burdensome costs of retail space. In this way, the shared model supports equitable participation in the urban food economy.

3. Social impact

Environmental management is a critical component of the Shared Kitchen 4.0 infrastructure, particularly given the F&B sector's traditionally high environmental footprint. The project addresses this by integrating waste mitigation and environmental control systems within the building design. A centralized greywater treatment facility is installed at each kitchen cluster, which filters and treats effluents before discharging them into the municipal sewer system, thus complying with TCVN 14:2008 standards on industrial wastewater treatment. The treated water meets Category B standards (non-potable reuse), which can be recycled for non-consumptive purposes like sanitation and landscaping.

Solid waste segregation is also operationalized at the kitchen level. Each tenant is required to comply with a three-bin system (organic, recyclable, residual), with organic waste collected daily for composting or processed into biofuel through a third-party partnership with a certified waste-to-energy service provider. Furthermore, the facility employs a Building Energy Management System (BEMS) integrated with IoT devices that monitor real-time energy consumption and optimize cooling, lighting, and cooking utilities. These features help reduce greenhouse gas emissions and operational inefficiencies. The use of LED panels, inverter-based cold storage systems, and high-efficiency kitchen exhaust hoods reduces total energy consumption by an estimated 18–22% compared to conventional kitchens. In urban planning terms, the project demonstrates a prototype for low-carbon, high-efficiency food production zones in megacities like Ho Chi Minh City.

4. Sustainable development orientation

The future development of Shared Kitchen 4.0 aligns closely with both Vietnam's Green Growth Strategy 2021–2030 and the United Nations Sustainable Development Goals (SDGs). In particular, the project contributes directly to SDG 8 (Decent Work and Economic Growth), SDG 11 (Sustainable Cities and Communities), and SDG 12 (Responsible Consumption and Production). The business model adopts a phased sustainability roadmap that integrates technology, governance, and education for long-term viability.

In the immediate term (1–3 years), Shared Kitchen 4.0 will introduce a sustainability training curriculum for all tenants, covering topics like energy efficiency, responsible waste disposal, food loss prevention, and digital kitchen tracking. A centralized dashboard

will be made available to tenants to monitor their resource consumption and receive feedback on how to improve. In the medium term (3–5 years), the project aims to achieve third-party green certification—such as LOTUS (Vietnam Green Building Council) or EDGE (IFC)—for its building envelope and operational performance. Additionally, future expansions into adjacent urban districts or satellite cities (e.g., Thu Duc, Binh Duong, or Long An) will incorporate modular green building elements like rooftop solar arrays and rainwater harvesting systems to further enhance environmental impact.

Over the long term (5–10 years), the project envisions transitioning into a circular economy hub. Initiatives will include collaboration with agri-tech startups to utilize food waste for urban farming, building a shared logistics network to reduce last-mile emissions, and establishing traceability systems using blockchain to ensure supply chain transparency. Furthermore, the project will explore public-private partnerships with local governments and development agencies to promote policy frameworks for shared-use F&B infrastructure, with potential for tax incentives or green bond financing. These efforts position Shared Kitchen 4.0 not only as a scalable business model, but also as a key player in shaping a more inclusive, resilient, and sustainable urban food ecosystem in Vietnam.

VII. CONCLUSION & RECOMMENDATIONS

1. Conclusion

In conclusion, Shared Kitchen 4.0 presents a viable and scalable business model that addresses the rising demand for affordable, regulation-compliant, and professionally equipped kitchen spaces in urban Vietnam. Located strategically in Tan Phu District, the project benefits from strong access to residential zones and delivery networks, while its modular design allows for operational flexibility and future expansion. With a sound financial structure, full legal compliance, and alignment with national policies supporting SME growth and food safety, Shared Kitchen 4.0 offers a practical, cost-efficient solution for food startups and small-scale F&B enterprises. As a well-prepared and forward-looking initiative, it is well-positioned to become a pioneering platform in Vietnam's emerging shared culinary infrastructure sector.

Backed by detailed financial modeling, the project exhibits a positive Net Present Value (NPV), competitive Internal Rate of Return (IRR), and an acceptable Payback Period under moderate and optimistic market scenarios. The use of quantitative risk analyses—one-way, two-way, scenario planning, and Monte Carlo simulation—reinforces confidence in the project's performance across a wide range of real-world uncertainties.

From a financial standpoint, the project achieves operational profitability by Year 3, with net cash flows stabilizing and increasing year over year. Sensitivity tests highlight that pricing strategies, particularly for large and medium kitchen segments, and initial occupancy rate are critical success factors. Operational risks are manageable, with debt servicing capacity improving significantly after the early ramp-up phase. The project's multidimensional impacts—economic, social, and environmental—align with Vietnam's national policy priorities and global SDGs, reinforcing its relevance beyond pure financial metrics.

2. Recommendation

These recommendations outline a focused approach to quickly secure market share and build a robust financial foundation for Shared Kitchen 4.0. By driving early occupancy, optimizing costs, and locking in stable pricing, the project can achieve strong cash-flow performance while laying the groundwork for sustainable, long-term growth:

Prioritize First-Year Occupancy: The first year is critical for profitability. Launch an aggressive pre-opening marketing campaign across multiple channels (e.g. Facebook Ads, Google Ads, email) targeted at F&B startups, home chefs, and delivery vendors. Offer attractive move-in incentives—such as 10–15% off rents for contracts of six months or longer, plus complimentary usage hours in the first month. Secure early partnerships with delivery platforms (GrabFood, ShopeeFood) and logistics providers to streamline tenants' onboarding and boost retention.

Implement Segmented & Long-Term Pricing: Adopt a tiered pricing model for micro-kitchens, small, medium, and large spaces, with preferential rates for F&B SMEs that sign 12–24-month leases. Negotiate fixed-price land leases of at least 3–5 years, indexed annually to CPI, to stabilize cash flows and safeguard NPV against rising input and land-rental costs.

Streamline Management & Sales Costs: Automate scheduling, billing, and administrative tasks using low-code tools (e.g. Google Forms + Apps Script, Notion + AppSheet) and move all contracts, invoices, and food-safety records to the cloud to cut printing and storage expenses. Set clear KPIs for the sales and marketing teams and outsource non-core functions (printing, digital ads) to keep M&S costs under 160 million VND per year.

Build Financial Resilience: Prepare for negative cash flow in years one and two by maintaining a liquidity reserve of at least 0.3% of annual revenue to cover unexpected expenses. Structure loan disbursements around project milestones (site readiness, pilot completion, full-scale rollout) to reduce interest burdens and improve cash-flow management.

Forge Strategic Partnerships & Certifications: Partner with food-safety authorities and apply for recognized green-building certifications (LOTUS, EDGE) to enhance credibility, attract institutional investors, and meet ESG criteria. Promote your “green kitchen” credentials by implementing waste segregation, eco-friendly infrastructure, and prioritizing socially-driven tenants (e.g. women-led businesses).

Plan Flexible Expansion & Continuous Optimization: After piloting in Tân Phú, scale modularly into other central districts and satellite zones (Thủ Đức, Bình Dương) to optimize capital deployment. Review key metrics—ADSCR, BCR, occupancy, and receivables—monthly, and adjust pricing quarterly based on market trends to sustain an IRR above 12% and a positive NPV. Lock in equipment costs via long-term purchase orders for major kitchen assets to hedge against input-price volatility.

Implementing these measures will enable Shared Kitchen 4.0 to capitalize on initial momentum, minimize risk, and scale efficiently—ultimately positioning it as a leading, resilient shared-kitchen solution in Vietnam.